



Chapter Notes: The World of Metals and Non-metals

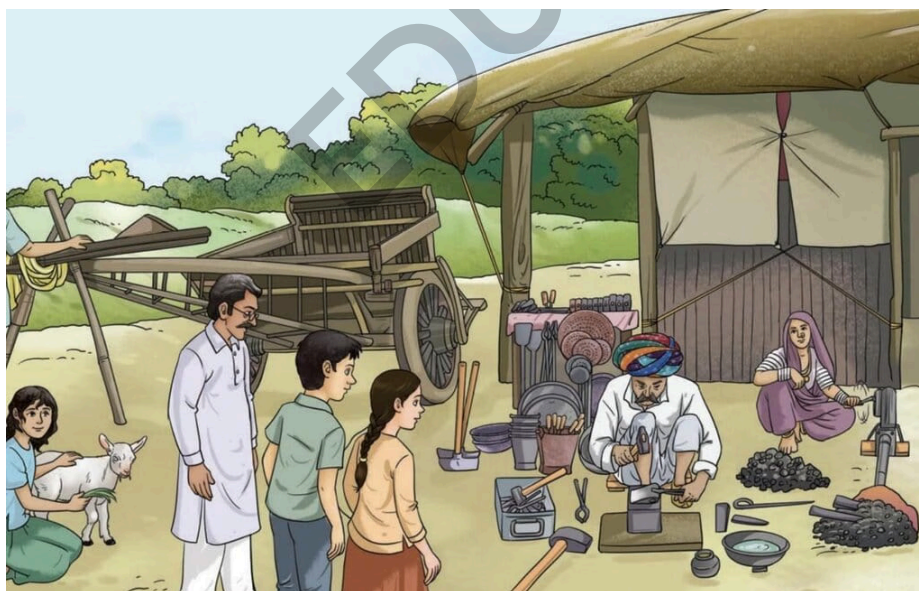
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Introduction

Yashwant and Anandi live in a village in Rajasthan. For their school project they decide to learn about ironsmiths - people who make useful items from metals. With their grandfather's help they visit Sudarshan uncle, a local ironsmith, to see how he shapes iron into everyday tools such as pans, buckets and farming equipment.



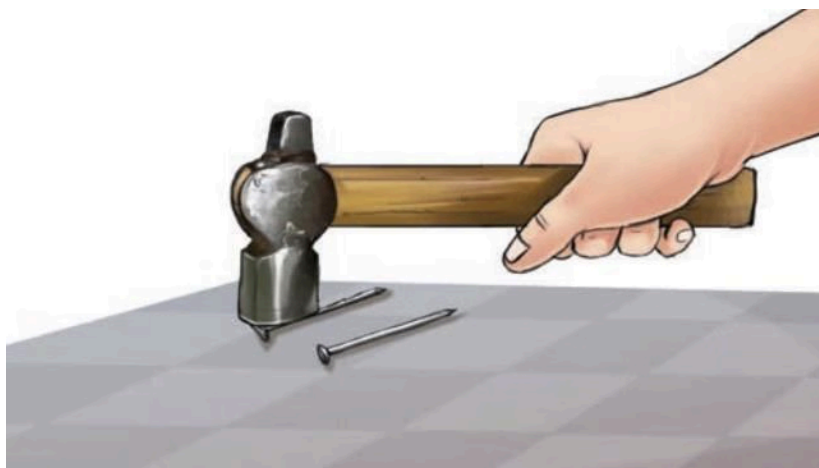
Curious and amazed, they begin exploring the fascinating world of metals, how metals differ from non-metals and the many ways these materials are used in daily life and in industry.

Properties of Materials

Materials are identified and classified by their properties - observable or measurable characteristics that help us group substances as metals or non-metals. Important properties used for classification include appearance, hardness, malleability, ductility, sonority, and the ability to conduct heat and electricity. Below we study these properties and compare how metals and non-metals behave.

Malleability

Malleability is the property of a material that allows it to be beaten or rolled into thin sheets without breaking.



Beating an iron nail with a hammer

Malleability of Metals

Most metals such as **copper**, **aluminium** and **iron** are malleable; they can be flattened into sheets or shaped by hammering and rolling.

Malleability is useful in everyday life: **silver foil** used to decorate sweets is made by hammering silver into extremely thin sheets, and **aluminium foil** used for wrapping food comes from aluminium's malleability.

Certain metals such as **gold** and **silver** are exceptionally malleable and can be made into very thin sheets called **gold leaf** and **silver leaf**.

Some exceptions and special cases

Not all metals are hard; metals like **sodium** and **potassium** are very soft and can be cut with a knife. They are much softer than copper or iron.

Mercury is a metal that is liquid at room temperature and therefore behaves differently from most metals (useful in devices such as thermometers and barometers).

Q: How does malleability benefit everyday life?

Malleability allows materials such as aluminium and silver to be formed into thin sheets and useful products like foil and decorative sheets.

Brittleness in Non-Metals

Non-metals such as coal and sulphur do not flatten on hammering; instead they tend to break or shatter into pieces. This characteristic is called **brittleness**.

Brittle materials cannot be bent or shaped easily; they break under pressure or impact.



Properties of Wood

Wood does not flatten like metals when hit with a hammer.

Wood neither flattens into sheets nor shatters like brittle substances; it shows a combination of flexibility and toughness.

Thus wood is generally considered neither malleable nor brittle in the strict sense used for metals and non-metals.

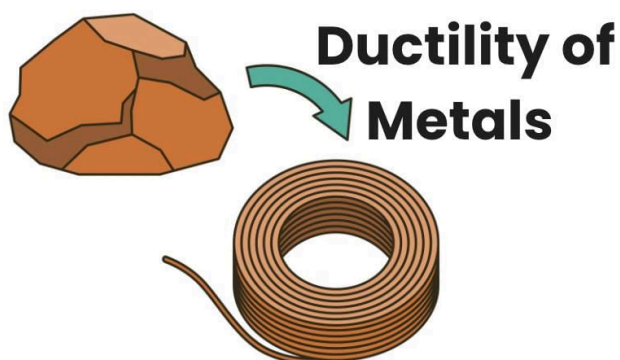
Ductility

Ductility is the property of a material that allows it to be drawn out into thin wires without breaking.

Ductility of Metals

Many metals, for example **copper** and **aluminium**, are ductile and can be drawn into wires. These wires are used extensively in electrical fittings, and to make ornaments like bangles and necklaces.

Gold is highly ductile; one gram of gold can be drawn into a wire several kilometres long under suitable conditions.



Everyday uses of ductility

Electrical fittings: Copper and aluminium wires are widely used in electrical wiring for homes and appliances because they conduct electricity well.



Jewellery and ornaments: Wires of gold, silver and other metals are shaped into bangles, necklaces and earrings.



Musical instruments: Metal strings are used in instruments such as the veena, sitar, violin and guitar; the strings vibrate to produce musical sound.



Ductility of Non-metals

Non-metals are generally not ductile; they break when stretched and cannot be drawn into wires.

Examples: **coal**, **sulphur** and **phosphorus** break into pieces when we try to stretch or hammer them.

Such non-metals are brittle rather than ductile.



Dive deeper - steel wires: **Steel** is an alloy of **iron** (a metal) and **carbon** (a non-metal). Steel wires and ropes are very strong and can support heavy loads; they are used in structures such as suspension bridges and in cranes to lift heavy objects safely.

Q: What does ductility refer to in materials science?

Ductility refers to a material's ability to deform under tensile stress, often characterised by its capacity to be stretched into a wire.

Sonority

Sonority is the ability of a material to produce a clear ringing sound when struck.

Metals are sonorous - objects such as spoons or coins give a ringing sound when struck. This property is why bells, ghungroos and many musical percussion instruments are made of metal.

Non-metals like coal and wood produce dull sounds when dropped or struck and are not sonorous.

Everyday examples of sonority

The ringing sound of **ghungroos** (small bells worn by dancers) is due to the sonorous nature of the metal pieces used.



The loud ringing of a **school bell** is produced by the sonority of the metal from which it is made.

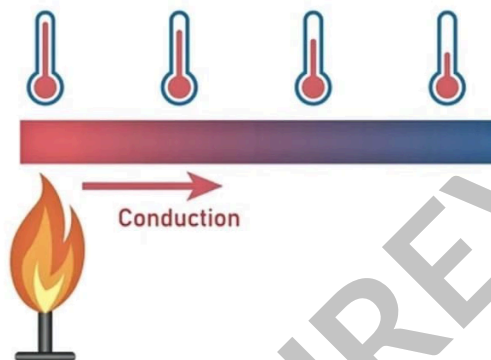


Sound differences from striking wood versus metal can help identify materials and are used in everyday navigation and tasks.



Conduction of Heat

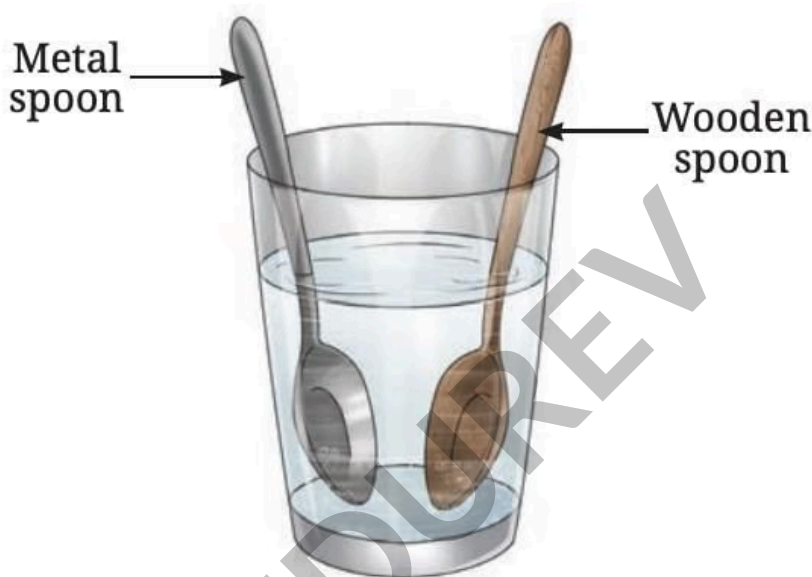
Conduction is the transfer of heat energy through a material from the hotter part to the cooler part.



Metals (for example, aluminium, copper and iron) are good conductors of heat and are commonly used for cooking vessels because they transfer heat quickly and evenly.

Non-metals such as wood are poor conductors of heat (they are insulators) and therefore remain cooler when placed in hot water; this is why vessel handles are often made of wood or plastic.

If a metal spoon and a wooden spoon are left in hot water for the same time, the metal spoon feels much hotter because metal conducts heat faster from the hot water to your hand while wood does not.



Good and poor conductors

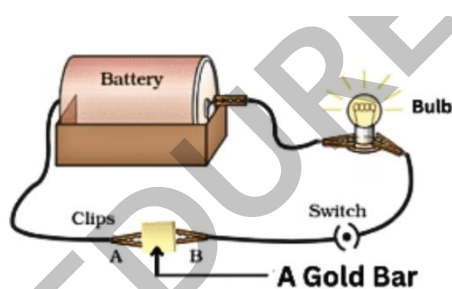
Materials that allow heat to pass through them easily are called **conductors**; metals are good conductors.

Materials that do not allow heat to pass through easily are called **insulators** or poor conductors; wood, plastic and rubber are typical insulators used for handles and safety purposes.

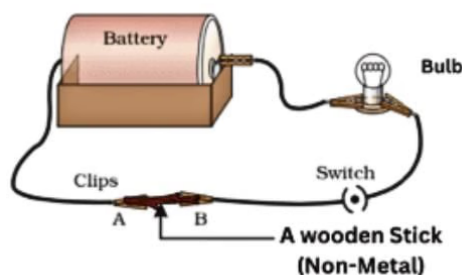
Conduction of Electricity

Materials that allow electric current to pass through them easily are called **good conductors of electricity**; those that do not are called **insulators**.

Metals such as aluminium, iron and copper are excellent conductors and are used for electrical wiring and many electrical devices.



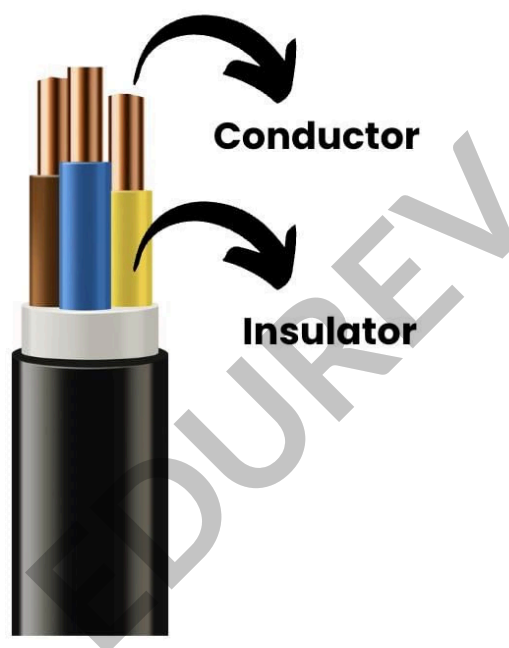
Non-metals such as sulphur, coal, wood, stone, rubber and nylon are poor conductors of electricity and are used as insulating materials (for example, in tool handles, cable coverings, and protective clothing).



Electrical conductivity and safety

For example, the handle of a **screwdriver** used by electricians is typically made of plastic or rubber because these are poor conductors of electricity and help protect the user from electric shock.

Electricians also wear rubber gloves and rubber-soled shoes to reduce the risk of electrical accidents since rubber does not allow electricity to pass through easily.



Metallic Lustre

Many metals (for example copper, aluminium and iron in polished form) show a characteristic shiny appearance called **metallic lustre**; they reflect light well and appear bright and polished.



In contrast, non-metals such as coal, sulphur and wood are generally dull (non-lustrous) and do not reflect light the way metals do.



Metals are often harder than many non-metals and resist scratching or denting under normal conditions.

Holistic lens - the impact of iron on the progress of Indian civilisation: In ancient India, early civilisations used metals such as copper and gold for tools and jewellery. The widespread use of **iron** came later. Iron's strength and durability made iron tools, especially agricultural implements like ploughs, very important for improving farming and contributing to the progress of civilisation.



Why was copper discovered before iron? Copper was discovered and used earlier because it occurs naturally in a pure form and has a lower melting point, making it easier to extract and work with using early human technology. Iron usually occurs as ores and requires higher temperatures and more advanced techniques to extract and shape.

MULTIPLE CHOICE QUESTION

Try yourself: What is the property of materials that allows them to be drawn into thin wires?

A Conductivity

B Malleability

C Ductility

D Sonority

[View Solution](#)

Effect of Air and Water on Metals: Iron

Rusting of iron is the formation of reddish-brown flaky deposits on iron surfaces when iron reacts with oxygen and moisture in the air. This chemical change is commonly called rusting.



Rusting Iron

The brown deposits that form on iron are known as **rust**.

Rusting is a chemical reaction in which iron reacts with oxygen in the presence of water (moisture).

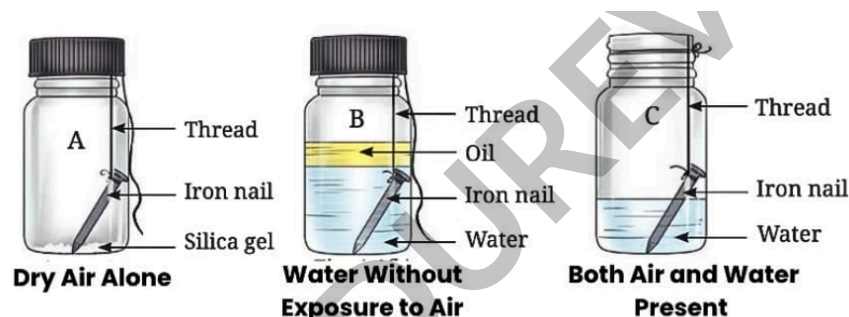
Rusting causes the iron to deteriorate and weaken over time, reducing the life and strength of iron objects.

Conditions required for rusting

Rusting occurs only when iron is exposed to both **air (oxygen)** and **water (moisture)** - that is, moist air.

Iron does not rust if exposed to dry air alone (no moisture).

Iron also does not rust when completely submerged in water that has no dissolved oxygen; the presence of oxygen is essential.



Impact

Rusting makes iron objects and structures weak and unsafe.

It causes damage and decay of iron used in buildings, vehicles, bridges and tools.

Large sums are spent every year on repairing or replacing rusted iron structures and equipment.

Prevention of rusting

Painting iron surfaces to keep air and moisture away.

Applying **oil or grease** to form a protective barrier on the metal surface.

Galvanisation: coating iron with a layer of **zinc** which protects iron from rusting.

Q: What role do air and water play in the corrosion of iron?

Air (oxygen) and water (moisture) together cause oxidation of iron; this oxidation process leads to the corrosion (rusting) of iron.

Corrosion

Corrosion is the gradual deterioration of metals due to chemical reactions with substances in the environment such as air, water, acids or salts. Rusting is a common example of corrosion for iron.

Fascinating fact - The Iron Pillar of Delhi: The Iron Pillar of Delhi, built over 1600 years ago during the time of Chandragupta II, is about 8 metres tall and weighs more than 6000 kilograms. Despite centuries of exposure to rain, wind and weather, it has hardly rusted, showing the high skill of ancient Indian metallurgists.



Effect of Air and Water on Other Metals

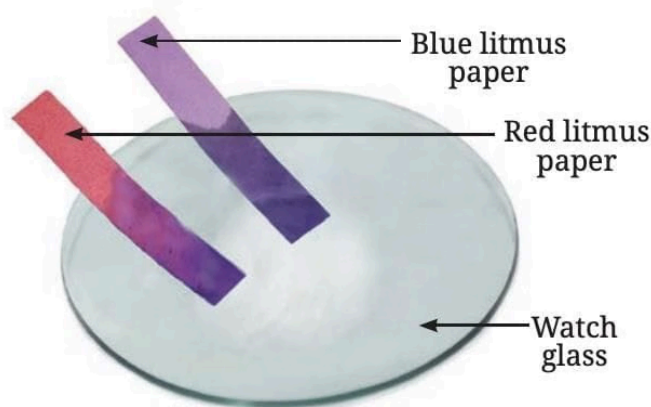
Magnesium

When a magnesium ribbon is burned it produces a very bright white flame and leaves a white powder, **magnesium oxide**.

Magnesium oxide reacts with water to form a solution that turns red litmus blue, indicating that magnesium oxide is **basic** in nature.

As a general rule, oxides of metals are often basic.





Sodium

Sodium reacts vigorously with air and water, producing heat and products such as sodium oxide or sodium hydroxide. For safety, sodium metal is stored under **kerosene** to prevent contact with air and moisture.

Sodium oxide is basic, consistent with the general behaviour of metal oxides.

MULTIPLE CHOICE QUESTION

Try yourself: What causes rust to form on iron objects?

- A Exposure to sunlight
- B Exposure to water only
- C Exposure to both air and water
- D Exposure to air only

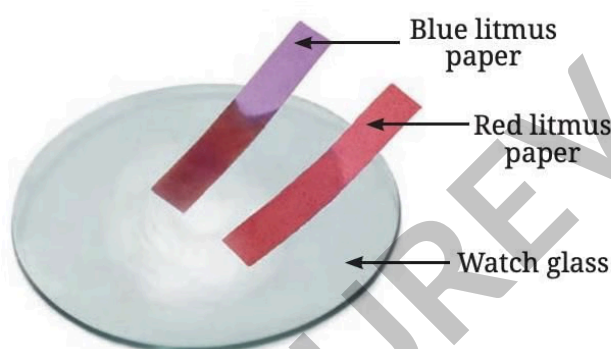
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Substances that Behave Differently from Metals in Air and Water

Certain substances, for example **sulphur** and **phosphorus**, behave very differently from metals on exposure to air and water.

Sulphur

When sulphur is burned, it produces **sulphur dioxide (SO_2)** gas. When sulphur dioxide dissolves in water, it forms **sulphurous acid**, which is acidic and turns blue litmus red. Sulphur itself does not react with water when simply mixed; its combustion products (such as SO_2) are responsible for acidic behaviour in solution.



Phosphorus

Phosphorus catches fire readily in air and therefore is usually stored under **water** to prevent it coming into contact with oxygen.

Properties of Non-metals

Non-metals are usually soft and dull (non-lustrous).

They are not malleable or ductile and therefore cannot be made into sheets or wires.

Non-metals are not sonorous; they produce dull sounds when struck.

They are poor conductors of heat and electricity (insulators).

Oxides of non-metals are generally **acidic** (for example, sulphur dioxide forms acidic solutions).

Examples of non-metal elements: **oxygen, hydrogen, nitrogen, carbon, chlorine, iodine.**

Materials such as plastic, glass, wood, rubber and paper are not elements and so are not classified as metals or non-metals in the elemental sense.

Dive deeper - Elements (the building blocks of matter):

Elements are pure substances that cannot be broken down into simpler substances by ordinary chemical methods.

Everything around us is made up of elements.

There are currently **118 known elements**.

Some elements occur naturally (for example oxygen, iron and gold); others are artificially created in laboratories.

Metals and non-metals are two important sub-categories of elements.

Are Non-Metals Essential in Everyday Life?

Although metals are highly visible due to their shine, strength and conductive properties, non-metals perform essential roles that keep life and industry functioning.

Importance of some non-metals

Oxygen: We breathe oxygen every day; it is essential for life and is also used in hospitals and in combustion and welding processes.

Carbon: Carbon is the building block of life - it forms the basis of proteins, fats, and carbohydrates which provide energy and enable growth.

Nitrogen: Widely used in the manufacture of fertilisers and chemicals; nitrogen is a vital nutrient for plant growth.

Chlorine: Used in water purification to make drinking water safe.

Iodine: Used as an antiseptic (iodine solution) on wounds to prevent infection.

Science and society

Many metals and **alloys** (mixtures of two or more metals, or metals with non-metals) are used daily in utensils, tools and machines. Metals are essential in modern technology and almost every industry. Examples include the use of **zirconium** in atomic energy and **titanium** in aerospace engineering. In India, recycling metals such as iron and aluminium is common and helps reduce waste and promote sustainability.

MULTIPLE CHOICE QUESTION

Try yourself: What happens when sulfur is burned in air?

- A It becomes a metal.
- B It turns blue litmus red.
- C It produces sulfur dioxide gas.
- D It reacts with water.

View Solution

Points to Remember

Metals are generally shiny, hard, malleable (can be beaten into sheets), ductile (can be drawn into wires), sonorous (produce ringing sounds) and good conductors of heat and electricity. Non-metals are usually dull, soft or brittle, and poor conductors of heat and electricity. Malleability allows metals such as copper and gold to be made into thin sheets used as foils and in tools. Ductility allows metals to be drawn into wires used in electrical systems and jewellery. Metals produce clear ringing sounds (used in bells); non-metals produce dull sounds.

Metals conduct heat and are used for cooking pots; non-metals such as wood are used for handles because they are poor conductors.

Metals such as copper and aluminium conduct electricity, while materials like rubber and plastic act as insulators for safety.

Iron rusts when exposed to both air and water, forming reddish-brown rust.

Copper and silver also form surface coatings (patina) when exposed to air and moisture (green or black coatings respectively).

The Iron Pillar of Delhi (over 1600 years old) demonstrates how ancient Indian metallurgists produced iron that resists rust.

Magnesium burns in air to form a basic oxide; sodium must be stored in kerosene to prevent rapid reaction with air.

Sulphur burns to form sulphur dioxide which forms acidic solutions; sulphur itself does not react with water.

Phosphorus is stored under water to prevent it catching fire in air; oxides of non-metals are generally acidic.

Non-metals like oxygen, carbon and nitrogen are vital for life, agriculture and industry.

There are 118 known elements, classified as metals and non-metals; some occur naturally and some are made in laboratories.

Metals and alloys are widely used in tools, machines and industries; recycling metals helps protect the environment.

Q: What properties of metals allow them to be used in making electrical wires and tools?

Malleability and ductility are the properties that allow metals to be shaped into tools and drawn into wires used in electrical applications.

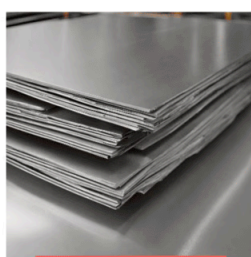
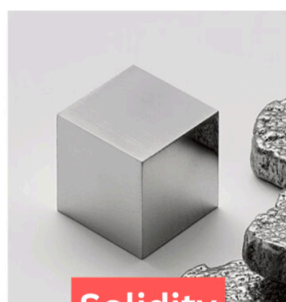
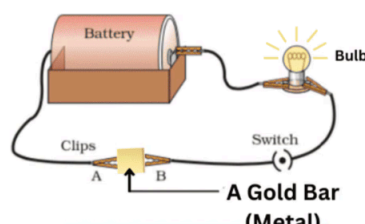
Difficult Words and Their Meanings

Malleability: The ability of a material to be beaten into thin sheets without breaking.

Ductility: The ability of a material to be drawn into thin wires.

Sonority: The property of a material to produce a ringing sound when struck.

Conduction: The transfer of heat or electricity through a material (for example, heat transfer through a metal spoon).

**Ductility****Malleability****Lustre****Solidity****Good Conductor
of Electricity**

Metal Characteristics

Conductor: A material, such as copper, that allows heat or electricity to flow easily.

Insulator: A material, such as rubber, that does not allow heat or electricity to flow easily.

Rusting: The process where iron forms brown deposits (rust) when exposed to both air and water.

Corrosion: Gradual damage to a metal's surface by air, water or other substances (for example, rust on iron or green coating on copper).

Oxide: A substance formed when a metal or non-metal reacts with oxygen (for example, magnesium oxide, sulphur dioxide).

Element: A basic substance, such as iron or oxygen, that cannot be broken into simpler substances by ordinary chemical methods; 118 known elements.

MULTIPLE CHOICE QUESTION

Try yourself: Which property is NOT characteristic of metals?

A Dull appearance

B Sonorous

C Good conductivity

D Malleable

View Solution

You can practice questions from this chapter here: **Very Short Question Answer: The World of Metals and Non-metals** (https://edurev.in/studytube/Class-7-Science-Chapter-4-Very-Short-Question-Answer--The-World-of-Metals-and-Non-metals/b8bd319f-784a-4813-9c42-6c672daf1b26_t)